

flexible monitoring across hybrid and multi-cloud environments.

## Integrating New Relic with Dockerized applications

Due to space constraints, images for all the integration steps have not been given here. Interested readers can visit the GitHub repository for the detailed report and project at <https://github.com/rishithaiitb/ImageCaptioning.git>.

The image captioning system was developed and deployed by integrating a machine learning model into the application, following DevOps best practices. This system generates descriptive text for images, enhancing accessibility for visually impaired individuals, automating content creation, and improving search engine optimisation. Following its successful deployment, New Relic was integrated to enable real-time monitoring and performance optimisation.

## Setting up a New Relic account and licence key

Before integrating New Relic with a Dockerized application, it is essential to ensure that the application is fully containerised and running smoothly within Docker. Additionally, the containers must be configured with the appropriate network settings to allow seamless communication between services.

The first step in the integration process is to create a New Relic account and obtain a licence key, which is required to establish a connection between the New Relic agent and the New Relic platform for collecting application performance data. Here are the steps for that.

**Create a New Relic account:** Visit the New Relic *Signup Page* to register for an account. Provide the necessary details and complete the signup process. Once registered, log in to your New Relic dashboard.

**Access the New Relic dashboard:** After logging in, navigate to the New Relic dashboard, where all monitoring entities and configurations are available.

**Generate a licence key:** Go to the *API Keys* section in your profile settings. Navigate to the *License Key* tab and select *Create a New License Key*. Copy the generated licence key, as it will be required for agent configuration in later steps.

This licence key acts as an authentication token that allows the New Relic agent to collect real-time application performance data and send it securely to the monitoring dashboard.

## Java agent for backend monitoring

The backend of the application is built using Spring Boot and Java running within the Docker container. To ensure real-time performance monitoring and proactive issue detection, the New Relic Java agent is integrated into the application.

This agent provides detailed performance insights, including:

- Response times, error rates, and throughput
- Database query performance and execution times
- JVM metrics such as garbage collection, heap memory, and thread activity
- External service interactions and API response times

By monitoring these critical metrics, the New Relic Java agent helps identify performance bottlenecks, optimises resource utilisation, and enhances application reliability. Here is a step-by-step guide for integrating the New Relic Java agent with a Spring Boot backend running in Docker.

### Step 1: Select the Java Agent in New Relic

- Navigate to the *Integrations & Agents* section in New Relic.
- Choose *Java* as the integration option.

### Step 2: Configure the licence key

- Select the environment where the application is running (in this case, Docker).
- Enter the New Relic licence key generated earlier.
- Click *Continue* to proceed.

### Step 3: Choose the operating system

- Specify the operating system of the application (for this setup, select Linux).

### Step 4: Download and install the Java agent

- Download the Java agent using the following command:

```
curl -O https://download.newrelic.com/newrelic/java-agent/newrelic-agent/current/newrelic-java.zip
```

- Extract the downloaded package.

### Step 5: Configure the Java agent

- Assign a custom application name in the New Relic dashboard.
- Download the New Relic configuration file containing your API key.
- Place this file inside the extracted Java agent directory.

### Step 6: Set environment variables in Docker

To allow the New Relic agent to collect metrics, modify the Docker and Docker Compose files by setting environment variables:

```
ENV NEW_RELIC_LICENSE_KEY= YOUR LICENSE KEY
ENV NEW_RELIC_APP_NAME=image-captioning-backend
```

### Step 7: Update the Dockerfile

- Add the New Relic Java agent to the Dockerfile.